0580_w24_ms_42_Q10

Question	Answer	Marks	Partial Marks	Guidance
10(a)	$3x^2 - 18x - 48$ final answer	**2**	**B1** for two correct terms or for correct answer seen then spoiled	Accept equivalents for B2 or B1 e.g. B2 for $3(x^2 - 6x - 16)$ Condone poor notation for B2 or B1 e.g. $y = 3x^2 - 18x - 48$, $y = 3x^2 - 18x$
10(b)	*their* $\frac{dy}{dx} = 0$ soi	**M1**		After 2 marks in (a) then must use correct derivative = 0 to score M1 M1 can be implied by a correct method without errors to find x or for answers $x = -2$ and $x = 8$ FT *their* dy/dx
	$[3](x - 8)(x + 2)$ oe or $\frac{-(-18) \pm \sqrt{(-18)^2 - 4(3)(-48)}}{2 \times 3}$ oe or $3 \pm \sqrt{16 + 3^2}$	**M1**	oe $\frac{18 \pm \sqrt{900}}{6}$	FT *their* 3-term quadratic oe eg $(3x - 24)(x + 2)$ or $(3x + 6)(x - 8)$ Allow B1 for correct $(ax + b)(cx + d)$ from *their* quadratic Allow 18^2 and allow -18^2 if recovered. Allow 18 for $-(-18)$ **No recovery for short line or short root unless seen or implied in working** Fraction line must underscore the ± sign Length of root must cover – sign, ie reach 4 or complete square method correct for *their* quadratic
	(–2, 52) (8, –448)	**B2**	**B1** for one correct pair of coordinates or for two correct values of x	

Question	Answer	Marks	Partial Marks	Guidance
10(c)	(–2, 52) maximum with reason and $(8, -448)$ minimum with reason and no incorrect statement	**3**	Reasons could be eg 1. A reasonable sketch of a positive cubic 2. Correct evaluation and use of 2nd derivative $6x - 18 = -30$, $-30 < 0$, so (–2, 52) is a maximum oe. $6x - 18 = 30$, $30 > 0$, so $(8, -448)$ is a minimum oe. 3. Evaluates correctly values of y on both sides of both correct stationary points 4. Finds gradient on each side of both correct stationary points. Any incorrect statement MAX B2 **B2** for 1 correct with correct reason for that stationary point or for **both** x-values correct and reasonable sketch of a positive cubic or for correct substitution and evaluation of **both** of *their* x-values into *their* second derivative or substitution and evaluation for one x-value on both sides of **both** of *their* stationary points to find the gradients soi or **M1** for showing [2nd derivative =] $6x - 18$ or correct FT *their* 2nd derivative from part (a) or substitution and evaluation shown for one x-value on both sides of **one** of *their* stationary points to find the gradients soi or for sketch of any positive cubic.	For 3 marks (or 2 marks) Allow (–2, 52) and $(8, -448)$ to be implied from working if e.g. using 2nd derivative = $6x - 18$, $6(-2) - 18 = -30$ so negative, maximum $6(8) - 18 = 30$ so positive, minimum Condone poor notation for 2nd derivative For **B3** or **B2** accept reasonable sketch drawn in part (c) or referred to if drawn in part (b) accept without axes For **B3** if drawn on axes then max must be in 2nd quadrant and min must be in 4th quadrant For **B2** if drawn on axes allow for max in 2nd quadrant **or** min in 4th quadrant The second derivative must be correct FT from *their* first derivative for **B2** or **M1** May use substitution into ***their*** dy/dx with stated x - values Allow evaluation to 1sig fig rounded or truncated in this context May use substitution into ***their*** dy/dx See AG